

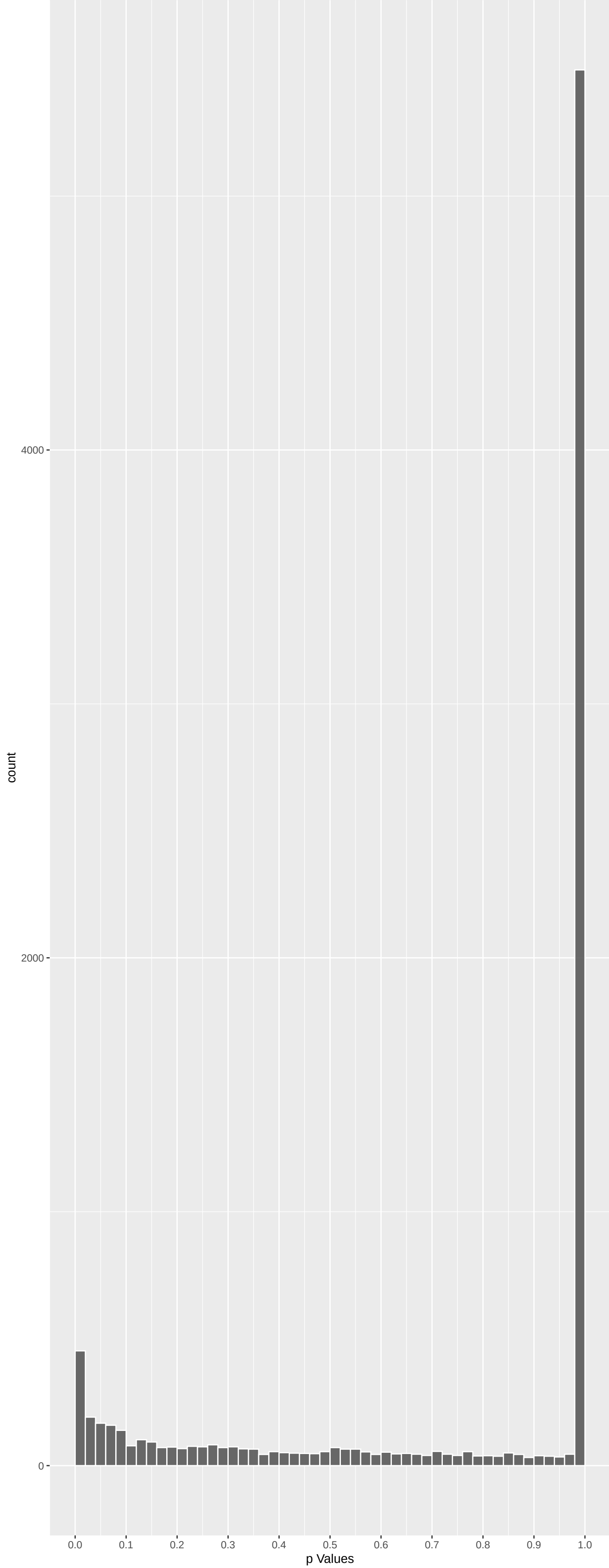
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
ADGRG4	-5.685340	7.833148e-08	1.183e-03	1.000e+00
PDE6B	5.368686	4.758744e-07	2.396e-03	1.000e+00
CAMTA2	-5.417002	3.636398e-07	2.396e-03	1.000e+00
TASOR2	-5.286983	7.461014e-07	2.520e-03	1.000e+00
ROS1	-5.266534	8.341416e-07	2.520e-03	1.000e+00
OR6Y1	-5.165133	1.441611e-06	3.629e-03	1.000e+00
NEK5	-5.091450	2.132010e-06	4.600e-03	1.000e+00
SUSD5	-4.947353	4.513768e-06	7.575e-03	1.000e+00
DYNC2H1	-4.947543	4.509375e-06	7.575e-03	1.000e+00
STAMBPL1	-4.729666	1.349336e-05	2.038e-02	1.000e+00
DNAH9	-4.699127	1.567653e-05	2.131e-02	1.000e+00
TRPM7	4.683350	1.693346e-05	2.131e-02	1.000e+00
CEP126	-4.660254	1.894921e-05	2.202e-02	1.000e+00
CLU	4.627432	2.221369e-05	2.397e-02	1.000e+00
TLR8	4.578712	2.807089e-05	2.601e-02	1.000e+00
GRIN3B	-4.569910	2.927597e-05	2.601e-02	1.000e+00
PHYH	-4.587921	2.686062e-05	2.601e-02	1.000e+00
HK2	4.541152	3.356862e-05	2.817e-02	1.000e+00
HIRA	4.517323	3.757580e-05	2.970e-02	1.000e+00
IQGAP3	-4.500256	4.072301e-05	2.970e-02	1.000e+00
MYRFL	-4.497325	4.128827e-05	2.970e-02	1.000e+00
DAZL	4.464632	4.812394e-05	3.095e-02	1.000e+00
DHX34	-4.473805	4.610398e-05	3.095e-02	1.000e+00
GOLGA4	-4.441591	5.357764e-05	3.095e-02	1.000e+00
PRUNE2	-4.448762	5.181990e-05	3.095e-02	1.000e+00

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	0.72348895	1.000000e+00	1.000e+00	1.000e+00
FKBP4	1.05560188	1.000000e+00	1.000e+00	1.000e+00
CYP2B1	-0.01449733	1.000000e+00	1.000e+00	1.000e+00
NDUFAF7	-0.98055438	1.000000e+00	1.000e+00	1.000e+00
FUCA2	-0.81018634	1.000000e+00	1.000e+00	1.000e+00
DBNDD1	1.32171878	1.000000e+00	1.000e+00	1.000e+00
HS3ST1	1.47680964	8.383600e-01	1.000e+00	1.000e+00
SEMA3F	0.92798317	1.000000e+00	1.000e+00	1.000e+00
CFTR	-0.55133307	1.000000e+00	1.000e+00	1.000e+00
CYP51A1	4.40590269	6.320655e-05	3.292e-02	1.000e+00
USP28	0.62375025	1.000000e+00	1.000e+00	1.000e+00
SLC7A2	1.12340570	1.000000e+00	1.000e+00	1.000e+00
HSPB6	0.62110295	1.000000e+00	1.000e+00	1.000e+00
PKA	2.02934543	2.545387e-01	1.000e+00	1.000e+00
USH1C	-0.68212255	1.000000e+00	1.000e+00	1.000e+00
BAIAP2L1	2.20081112	1.664964e-01	1.000e+00	1.000e+00
TMEM132A	-1.19247448	1.000000e+00	1.000e+00	1.000e+00
DVL2	-0.82859883	1.000000e+00	1.000e+00	1.000e+00
RPA3	-1.34604184	1.000000e+00	1.000e+00	1.000e+00
HOXA11	0.26716787	1.000000e+00	1.000e+00	1.000e+00
TRAPPC5A	-0.25649526	1.000000e+00	1.000e+00	1.000e+00
SPATA20	-1.28978880	1.000000e+00	1.000e+00	1.000e+00
CD79B	0.47635053	1.000000e+00	1.000e+00	1.000e+00
RHBDD2	-0.69089795	1.000000e+00	1.000e+00	1.000e+00
RPUSD1	1.72380856	5.084545e-01	1.000e+00	1.000e+00

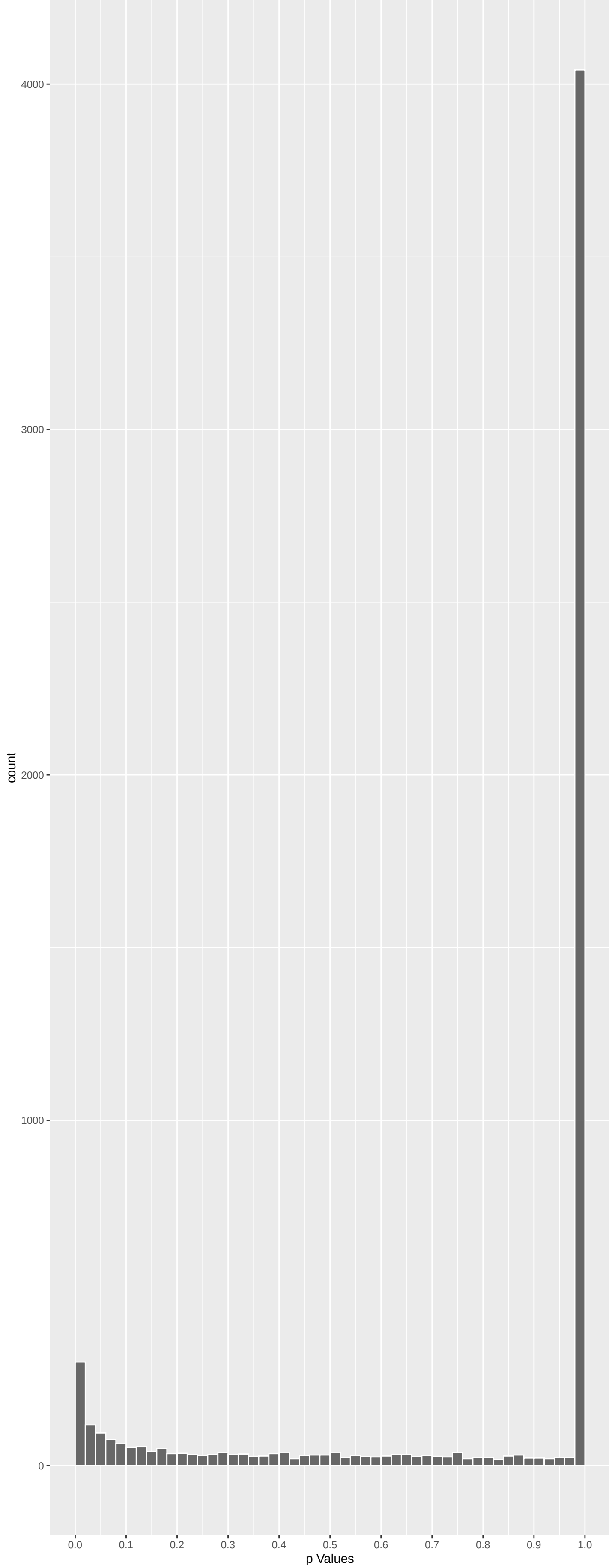
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
PDE6B	5.368686	4.758744e-07	2.396e-03	1.000e+00
TRPM7	4.683350	1.693346e-05	2.131e-02	1.000e+00
CLU	4.627432	2.221369e-05	2.397e-02	1.000e+00
TLR8	4.578712	2.807089e-05	2.601e-02	1.000e+00
HK2	4.541152	3.356862e-05	2.817e-02	1.000e+00
HIRA	4.517323	3.757580e-05	2.970e-02	1.000e+00
DAZL	4.464632	4.812394e-05	3.095e-02	1.000e+00
CYP51A1	4.405903	6.320655e-05	3.292e-02	1.000e+00
PIGG	4.361507	7.750192e-05	3.658e-02	1.000e+00
HERC2	4.267346	1.186876e-04	4.597e-02	1.000e+00

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
ADGRG4	-5.685340	7.833148e-08	1.183e-03	1.000e+00
CAMTA2	-5.417002	3.636398e-07	2.396e-03	1.000e+00
TASOR2	-5.286983	7.461014e-07	2.520e-03	1.000e+00
ROS1	-5.266534	8.341416e-07	2.520e-03	1.000e+00
OR6Y1	-5.165133	1.441611e-06	3.629e-03	1.000e+00
NEK5	-5.091450	2.132010e-06	4.600e-03	1.000e+00
SUSD5	-4.947353	4.513768e-06	7.575e-03	1.000e+00
DYNC2H1	-4.947543	4.509375e-06	7.575e-03	1.000e+00
STAMBPL1	-4.729666	1.349336e-05	2.038e-02	1.000e+00
DNAH9	-4.699127	1.567653e-05	2.131e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cilium Assembly (GO:0060271)	-0.07760309	197	1.877e-04	5.039e-01	CEP126:1 DYNC2H1:5 KIAA0753:28 GORAB:36 TEK1:92 DNAH5:138
Replication Fork Processing (GO:0031297)	-0.18268529	37	1.220e-04	5.039e-01	EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740 DNA2:772
DNA Repair (GO:0006261)	-0.06268710	234	1.031e-03	9.225e-01	USP45:10 XRC5:31 ATM:60 MLH1:66 MLIH:90 EME2:124
G Protein-Coupled Glutamate Receptor Sig	0.32713099	9	6.790e-04	9.225e-01	TRPM1:87 HOMER3:113 GRM5:318 GRM8:415 GRM1:445 GRM4:491
Double-Strand Break Repair (GO:0006302)	-0.08361591	134	8.687e-04	9.225e-01	ZMYND8:23 XRC5:31 ATM:60 EME2:124 RNF169:132 BRCA1:206
Regulation Of Cellular Componen	0.11602502	70	8.064e-04	9.225e-01	CLU:5 TGFBI:22 ATP10A:56 LRP1:67 BID:143 DCTN1:267
'De Novo' AMP Biosynthetic Process (GO:0	-0.10120681	4	4.833e-01	9.602e-01	PFAS:33 PAICS:6785 ATIC:6785 ADSS1:6785 NA NA
'De Novo' Post-Translational Protein Fol	0.12126361	15	1.040e-01	9.602e-01	SDF2L1:76 GAK:122 HSPAR:475 ENTDP5:5881 PTGES3:5881 DNAJB13:5881
2-Oxoglutarate Metabolic Process (GO:000	0.05634844	13	4.819e-01	9.602e-01	OGDHL:151 D2HGDH:960 IDH1:5881 GOT2:5881 GOT1:5881 IDH2:5881
3--UTR-mediated mRNA Destabilization (GO	0.07067574	11	1.171e-01	9.602e-01	MOV10:609 DHX36:916 RC3H1:5881 ZFP36L1:5881 ZFP36L2:5881 ZFP36:5881
3--UTR-mediated mRNA Stabilization (GO:0	0.10615460	11	1.229e-01	9.602e-01	DAZL:107 MAPK14:1866 BOLL:963 ZFP36:5881 TARDBP:5881 YBX3:5881
3'-Phosphoadenosine 5'-Phosphosulfate Me	-0.04012683	8	6.943e-01	9.602e-01	ENPP1:436 PAPSS2:1069 TPST2:6785 SULT2B1:6785 SULT1B1:6785 SULT1E1:6785
5S Class rRNA Transcription By RNA Polym	-0.05188503	5	6.890e-01	9.602e-01	GTF3C5:1509 GTF3C2:6785 GTF3C3:6785 GTF3C4:6785 GTF3C6:6785 NA
7-Methylguanosine RNA Capping (GO:000945	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA
7-Methylguanosine mRNA Capping (GO:00063	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA
ADP Transport (GO:0015866)	-0.11326131	5	3.889e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 NA
AMP Biosynthetic Process (GO:0006033)	-0.13424773	6	2.542e-01	9.602e-01	PFAS:33 APRT:864 PAICS:6785 ATIC:6785 ADSS1:6785 NUDT2:6785
AMP Metabolic Process (GO:0046167)	0.07641078	9	4.274e-01	9.602e-01	NTSE:365 AK9:1217 ADSS1:5881 NUDT2:5881 NTSC1A:5881 AMPD1:5881
ATF6-mediated Unfolded Protein Response	-0.05345055	5	6.789e-01	9.602e-01	DDIT3:1400 MBTPS1:6785 XBP1:6785 ATF6B:6785 CREBZF:6785 NA
ATP Metabolic Process (GO:0046034)	0.03454527	25	5.501e-01	9.602e-01	TGFBI:22 MYH7:163 NTSE:365 HSPA8:475 NUDT2:5881 ATP5P0:5881
ATP Synthesis Coupled Electron Transport	-0.11555397	3	4.882e-01	9.602e-01	NDUUF3:1179 NDUFB:6785 NDUFS2:6785 NA NA NA
ATP Transport (GO:0015867)	0.02195108	9	8.196e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 CALHM3:5881
AV Node Cell Acton Potential (GO:008601)	0.03753371	6	3.617e-01	9.602e-01	SCN10A:124 CACNB2:5881 SCN5A:5881 SCN4B:5881 CACNA1G:5881 TRPM4:5881
AV Node Cell To Bundle Of His Cell Commu	0.03089431	4	8.306e-01	9.602e-01	SCN5A:5881 CXADR:5881 GJC3:5881 GJA5:5881 NA NA
Arp2/3 Complex-Mediated Actin Nucleation	-0.19273439	10	3.485e-02	9.602e-01	WASHC1:500 ARPC5:662 WHAMM:1033 ARPCL5:1228 TRIM27:1940 ARPC1A:2019
B Cell Activation (GO:0042113)	0.05020335	61	1.757e-01	9.602e-01	HDAC9:183 HDAC4:242 CD86:253 CD40:389 DCAFI:400 FLT3:506
B Cell Activation Involved In Immune Res	0.03091495	12	1.708e-01	9.602e-01	MSH2:5881 GPR183:5881 CD40LG:5881 CD180:5881 LIG4:5881 RNF8:5881
B Cell Chemotaxis (GO:0035754)	-0.04914117	5	7.036e-01	9.602e-01	GAS6:1685 PIK3CD:6785 HSD3B7:6785 CXCL13:6785 CYP7B1:6785 NA
B Cell Differentiation (GO:0030181)	0.04131841	44	3.435e-01	9.602e-01	HDAC9:183 HDAC4:242 DCAFI:400 FLT3:506 PTD52:629 ZBTB7A:976
B Cell Mediated Immunity (GO:0019724)	-0.03123557	13	6.956e-01	9.602e-01	SLA2:1193 MLK:1200 IL9R:1688 CD70:6785 MUSB1:6785 FCGR2B:6785
B Cell Proliferation (GO:0042100)	0.08607361	15	2.485e-01	9.602e-01	CD40:389 MEF2C:979 IL10:5881 CD70:5881 IFNE:5881 PTPRC:5881
B Cell Receptor Signaling Pathway (GO:00	0.09317267	26	1.003e-01	9.602e-01	CDSH:188 ATG16L1:6785 VAV3:651 MEF2C:979 TEC:1073 SH2B2:1198
BMP Signaling Pathway (GO:0030509)	-0.06775707	42	1.291e-01	9.602e-01	SLC39A5:455 LEFTY1:659 RUNX2:710 GDF9:795 INHBE:813 INHCB:856
C-terminal Protein Amino Acid Modificati	-0.20084384	8	4.919e-02	9.602e-01	GPLD1:188 AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 ATG16L1:6785
C-terminal Protein Deglutamylation (GO:0	-0.30359661	4	3.550e-02	9.602e-01	AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 NA NA
C-terminal Protein Lipidation (GO:000650	-0.14089015	3	3.980e-01	9.602e-01	GPLD1:188 ATG16L1:6785 ATG7:6785 NA NA NA
CA-dicarboxylate transport (GO:0015174)	0.03726866	10	6.836e-01	9.602e-01	SLC25A12:46 SLC25A18:569 SLC25A13:5881 SLC13A2:5881 SLC1A1:5881 UCP2:5881
C4-positive, Alpha-Beta T Cell Activati	0.03089745	5	8.109e-01	9.602e-01	STOML2:5881 TNFSF4:5881 RSAD2:5881 NKG7:5881 HMG6B1:5881 NA
CD40 Signaling Pathway (GO:0023035)	0.03937031	7	1.836e-01	9.602e-01	CD86:253 CD40LG:5881 PHB2:5881 TNIP2:5881 ITGAS:5881 RNF31:5881
CENP-A Containing Chromatin Assembly (GO	-0.05491412	5	6.707e-01	9.602e-01	HJURP:1304 MIS18A:6785 CENPI:6785 OIP5:6785 NASP:6785 NA

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.26593169	27	1.741e-06	1.128e-02	PBXIP1:32 ADAM15:55 RUSC1:113 MTX1:253 DCST2:335 THBS3:483
MIKKELSEN_ES_ICP_WITH_H3K4ME3	-0.05364367	532	2.757e-05	3.780e-02	SLC2A10:9 ZMYND8:23 PBXIP1:32 ZBTB4:42 ANXA3:48 WDCCP:72
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	-0.11506474	111	2.918e-05	3.780e-02	XRC5:31 ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419
REACTOME_MEIOTIC_RECOMBINATION	-0.22184976	31	1.926e-05	3.780e-02	ATM:60 MLH1:66 MLIH:90 BRCA1:206 RAD50:740 H2AX:792
REACTOME_DISEASES_OF_DNA_REPAIR	-0.17815757	47	2.413e-05	3.780e-02	ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTHYH:211 HUS1:459
REACTOME_DEVELOPMENTAL_BIOLOGY	0.04582905	669	6.278e-05	5.816e-02	CACNA1S:17 TGFBI:22 CNTN1:26 CDK8:31 PPARGA:34 DEK:54
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.05420268	466	7.010e-05	5.816e-02	COL1A2:9 TGFBI:22 IL33:24 TNFRSF4:39 RELB:47 IL1RL2:121
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.08802201	33	8.314e-05	5.816e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUSB1:419 RAD50:740
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	-0.11467024	98	8.981e-05	5.816e-02	XRC5:31 ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTHYH:211
WP_CILIOPATHIES	-0.09401272	148	8.217e-05	5.816e-02	DYNC2H1:5 KIAA0753:28 DDC2:192 SDCCAG8:200 DNAH11:217 CEP164:233
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	-0.06393525	310	1.180e-04	6.371e-02	ZMYND8:23 PBXIP1:32 WDCCP:72 DKM6B:180 ZNF608:216 CAVIN2:444
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	-0.14677250	58	1.121e-04	6.371e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HDR_THROUGH_SINGLE_STRAND_ANNEA	-0.19270778	32	1.599e-04	7.504e-02	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.06075307	330	1.622e-04	7.504e-02	SLC25A12:46 SLC20A1:49 NCKAP1:102 CHGB:117 ATP1A3:146 NPTN:192
REACTOME_INFECTIOUS_DISEASE	0.04764474	539	1.789e-04	7.722e-02	TLR8:10 ENTDP1:18 TGFBI:22 SHGL3:38 APB1:44 TUBB4A:59
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME	0.12260096	77	2.032e-04	8.132e-02	HIRA:3 DGC2R:53 DEFC05:60 ZDHHC8:130 PIKA:231 SERPIND1:250
KAUFFMANN_DNA_REPAIR_GENES	-0.08003677	181	2.135e-04	8.132e-02	XRC5:31 ATM:60 MLH1:66 MLIH:90 PMS2:169 BRCA1:206
REACTOME_HOMOLOG_DIRECTED_REPAIR	-0.11198819	89	2.659e-04	9.566e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HOMOLOGOUS_DNA_PAIRING_AND_STRA	-0.15910278	39	5.896e-04	1.909e-01	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
REACTOME_DNA_REPAIR	-0.06671598	226	5.768e-04	1.909e-01	USP45:10 XRC5:31 ATM:60 MLH1:66 EME2:124 PMS2:169
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.03884155	683	6.315e-04	1.947e-01	IWS1:7 COL1A2:9 TGFBI:22 SYMPK:23 CDK8:31 PPARGA:34
WP_INTERACTIONS_BETWEEN_IMMUNE_CELLS_AND	-0.20329116	23	7.390e-04	2.175e-01	TLR8:10 TGFBI:22 CD80:210 TGFBR2:234 CD86:253 CTLA4:625
WP_JOUBERT_SYNDROME	-0.12076775	64	8.438e-04	2.376e-01	ATM:60 CEP164:233 ARSL3B:363 MKS1:438 RAB8A:681 TCTN2:693
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.18559439	26	1.058e-03	2.854e-01	ATM:60 BRCA1:206 RAD50:740 DNA2:775 BRIP1:1030 WRN:1040
WINTER_HYPOXIA_METAGENE	0.06982545	176	1.443e-03	3.738e-01	TGFBI:22 PPARGA:34 SLC20A1:49 ENG:51 LRP1:67 HK2:182
CONRAD_STEM_CELL	0.17512172	27	1.640e-03	4.000e-01	CD9:70 DNMT3B:140 MYC:187 PIWIL2:251 DKK3:305 RET:321
WP_T_CELL_MODULATION_IN_PANCREATIC_CANCE	0.15148380	36	1.668e-03	4.000e-01	ENTPD1:18 TGFBI:22 TNFRSF4:39 CD80:210 CD86:253 NTSE:365
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	0.07281908	156	1.745e-03	4.035e-01	IL33:24 TNFSF10:78 RRM2B:89 CD163:114 BID:143 ITIH3:171
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.04240996	463	1.927e-03	4.160e-01	COL1A2:9 NFKBIE:32 HERC2:37 AP1B1:44 UBR2:50 TUBB4A:59
REACTOME_MEIOSIS	-0.12222087	54	1.907e-03	4.160e-01	ATM:60 MLH1:66 MLIH:90 BRCA1:206 STAG1:281 RAD50:740
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.06697460	180	2.003e-03	4.184e-01	TGFBI:22 TNFRSF4:39 TNFSF10:78 TGFBR2:234 TNFSF15:297 TNF:348
REACTOME_SIGNALING_OF_FUNCTION_OF_SMAPD2_3_IN	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAAD2:481 SMAAD4:5881 NA
REACTOME_SIGNALING_BY_TGF_BETA_RECEPTOR	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAAD2:481 SMAAD4:5881 NA
WP_SMCI5MC3_ROLE_IN_DNA_DAMAGE_CORNELIA	-0.31272267	8	1.912e-03	4.314e-01	ATM:60 BRCA1:206 RAD50:740 PAXIP1:1061 MRE11:1398 RAD18:2001
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05035964	312	2.349e-03	4.347e-01	COL1A2:9 TGFBI:22 IL33:24 IL1RL2:121 MYC:187 TRAF2:202
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_T	0.17889236	24	2.423e-03	4.358e-01	DEK:54 YY1:160 MYC:187 MYBL2:396 TFAP2D:529 TFAP2C:672
PID_FANCONI_PATHWAY	-0.13271960	43	2.671e-03	4.581e-01	ATM:60 BRCA1:206 FANCE:420 HUS1:459 RFCS:530 RAD50:740
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.14689116	35	2.701e-03	4.603e-01	SYMPK:23 SLU7:139 THOCS:198 LSM1:285 SLBP:409 CPSPF1:443
MIKHAYLOVA_OXIDATIVE_STRESS_RESPONSE_VIA	0.42563137	4	3.196e-03	4.813e-01	QAT:694 CTSF:761 AKR1B1:1136 PGAM1:1180 NA NA
KEGG_NON_HOMOLOGOUS_END_JOINING	-0.25695973	11	3.169e-03	4.813e-01	XRC5:31 RAD50:740 DNTT:1012 POLL:1047 XRC4:1239 NHEJ1:1351

DisGeNET Top pathways by non-permutation

Dermatitis, Allergic Contact	0.13501732	83	2.202e-05	4.300e-02	PPARGA:34 ACE:36 CNTN3:179 CD80:210 MRC1:224 HSD1B1:344
HIV Infections	0.05582680	511	2.045e-05	4.300e-02	TLR8:10 ENTDP1:18 TGFBI:22 GLUL28 TRPV1:45 CD9:70
Psoriasis	0.05514324	522	2.134e-05	4.300e-02	TLR8:10 HCAR2:19 TGFBI:22 IL33:24 PPARGA:34 ACE:36
Aortic Aneurysm, Abdominal	0.08501224	196	3.780e-05	6.152e-02	COL1A2:9 TGFBI:22 PPARGA:34 ACE:36 FCGR7:63 LRP1:67
Seminoma	0.09992233	135	6.476e-05	9.035e-02	CLU:5 ACE:36 ENTDP6:99 AFP:118 DNMT3B:140 GGLT3:177
Dermatitis	0.09661779	138	9.363e-05	1.046e-01	TGFBI:22 IL33:24 PPARGA:34 TRPA1:40 KLU7:74 LPN2:110
Inflammatory dermatitis	0.10557842	115	9.636e-05	1.046e-01	TGFBI:22 IL33:24 PPARGA:34 TRPA1:40 LPN2:110 PTGDS:164
Myasthenia Gravis	0.10245538	120	1.112e-04	1.086e-01	TGFBI:22 TNFRSF4:39 DNMT3B:140 CD86:253 ITGA5:74 KRT5:277
Scleroderma	0.10185243	119	1.302e-04	1.156e-01	COL1A2:9 TGFBI:22 ENG:51 RTCA:94 YY1:160 COMP:169
Dermatitis, Atopic	0.06237939	316	1.563e-04	1.174e-01	TGFBI:22 IL33:24 PPARGA:34 RELB:47 KLU5:58 FCGR7:63
Eczema	0.06337804	306	1.556e-04	1.174e-01	TGFBI:22 IL33:24 PPARGA:34 KLU5:58 KLU7:74 TRPM1:87
Malignant neoplasm of stomach	0.03067075	1450	1.801e-04	1.256e-01	CLU:5 COL1A2:9 TGFBI:22 SYMPK:23 IL33:24 CNTN1:26
Lymphoma, T-Cell, Cutaneous	0.08092886	177	2.182e-04	1.421e-01	TGFBI:22 TNFSF10:78 SCPEP1:128 HDAC9:183 MYC:187 TRAF2:202
Hepatocarcinogenesis	0.05865038	326	3.077e-04	1.878e-01	CLU:5 TGFBI:22 GLUL28 PPARGA:34 TRIM24:57 TNFSF10:78
Epilepsy, Temporal Lobe	0.09330872	122	3.874e-04	2.033e-01	GLUL28 ACE:36 TRPV1:45 DEPDC5:60 TNFSF10:78 SCN2A:145
Chronic Lymphocytic Leukemia	0.03985666	708	3.954e-04	2.033e-01	ENTPD1:18 TGFBI:22 NFKBIE:32 PPARGA:34 ACE:36 AQPI:42
Pemphigus Vulgaris	0.15896573	42	3.692e-04	2.033e-01	FCGR7:63 GJA8:120 DSG3:213 TNF:348 CD28:416 TNFRSF9:536
Ulcerative Colitis	0.04339224	585	4.169e-04	2.036e-01	CLU:5 TGFBI:22 IL33:24 PPARGA:34 ACE:36 HERC3:27
Cataract, Puvulent	0.44789038	5	5.237e-04	2.325e-01	GJA8:120 GJA3:238 MYCB1:741 HSF1:954 BFP5P2:1137 NA
Giant Cell Glioblastoma	0.12864985	61	5.204e-04	2.325e-01	CD9:70 TNFSF10:78 CRYBB1:741 NOTCH3:333 BLHE40:403 FNI:425
Tuberculosis, Pulmonary	0.09154807	117	6.489e-04	2.391e-01	TLR8:10 TGFBI:22 TRPV5:25 PDE6B:30 INTS4:69 CD9:70
Cold intolerance	0.40614217	6	6.710e-04	2.391e-01	PPARGA:34 TRPA1:40 TRPV1:45 UCP1:497 IFI44:560 LPIN1:5881
Crohn Disease	0.04203590	573	7.100e-04	2.391e-01	TLR8:10 ENTDP1:18 TGFBI:22 IL33:24 PPARGA:34 ACE:36
Huntington Disease	0.05482636	599	6.179e-04	2.391e-01	TGFBI:22 CNTN1:26 ACE:36 SHGL3:38 OPN1LW:105 CD163:114
Lymphoma	0.03520338	858	6.318e-04	2.391e-01	CLU:5 TGFBI:22 IL33:24 ACE:36 FCGR7:63 TNFSF10:78
Systemic Scleroderma	0.05342320	347	7.047e-04	2.391e-01	COL1A2:9 TLR8:10 TGFBI:22 IL33:24 ACE:36 TNFRSF4:39
Virus Diseases	0.04222004	571	6.864e-04	2.391e-01	TLR8:10 TGFBI:22 IL33:24 PPARGA:34 TNFRSF4:39 DEK:54
Carcinoma, Transitional Cell	0.06061514	260	8.319e-04	2.428e-01	TGFBI:22 PPARGA:34 ENG:51 CD9:70 TNFSF10:78 RRM2:89
Goldharsh Syndrome	-0.32012930	9	8.834e-04	2.428e-01	TSC1:186 FRAS1:295 GRIP1:296 TBC1D20:1220 TCOF1:1582 FGF8:1651
Hepatitis B	0.04377113	506	8.858e-04	2.428e-01	TLR8:10 TGFBI:22 PPARGA:34 ACE:36 DEPDC5:60 CD9:70
Herpes Simplex Infections	0.05406626	328	8.482e-04	2.428e-01	TGFBI:22 TRPV1:45 FCGR7:63 AFP:118 HSD1B1:344
Inflammatory disorder	0.07456169	143	8.745e-04	2.428e-01	TGFBI:22 IL33:24 TNFSF10:78 LPN2:110 HSD1B1:343
Leprosy	0.10265787	88	8.951e-04	2.428e-01	CD163:114 MRC1:224 TGFBR2:234 FAS:151 CTRP:38 CD40:389
Liver neoplasms	0.04343840	865	8.922e-04	2.428e-01	CLU:5 CACNA1S:127 TGFBI:22 GLUL28 PPARGA:34 TNFSF10:78
Multiple Myeloma	0.03471200	827	2.109e-04	2.428e-01	TGFBI:22 ENG:51 GRK6:64 LRP1:67 CD9:70 TNFSF10:78
Other specified connective tissue malformation	-0.31652030	9	1.011e-03	2.531e-01	KIF7:441 BBS4:807 MKS:925 LZTR1:1081 BBS2:120 BBS1:1293
Stomach Carcinoma	0.02710400	1428	1.066e-03	2.531e-01	CLU:5 COL1A2:9 TGFBI:22 IL33:24 CNTN1:26 GLUL28
Ciliopathies	-0.07751175	150	1.099e-03	2.617e-01	DYNC2H1:5 KIAA0753:28 DNAH5:138 DCCDC1:292 SDCCAG8:20 DNAH11:217